Instructions

1. The exam starts at **5:30 PM** and ends at **6:30 PM**.

2. Make sure that your student ID number and UW userid at the top of this page are correct!

3. Calculators are not permitted for this exam.

4. Answer the questions in the spaces provided. There are two blank pages at the end of the exam for extra space. **Clearly indicate what page extra work is on.**

5. Your exam paper will be scanned in greyscale, and only the scanned copy will be marked. Do not use highlighters, coloured pens or coloured pencils. Do not write too close to the margins.

6. Your answers must be stated in a clear and logical form. You must justify all of your answers (unless the question explicitly states that "no justifications required"). Your writing needs to be legible. Your arguments must be logical, clear and easy to understand. Simplify your answers as much as possible. You may use results from the lectures or homework without proof unless otherwise indicated. Clearly state which result you are using.

7. You may use the following formula for the inverse of a 2×2 matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

8. Good luck!

1) For the questions on this page, you are not required to justify your answer.

(a) 3 points – Consider the system

$$\begin{pmatrix} 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 5 & 4 & 3 \end{pmatrix} x = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$
$$x \geq \mathbb{0}.$$

Which of the following is a basic feasible solution for this LP? (Select exactly one answer.)

i. $(1, 0, 1, 0, 1, 0)^\top$
ii. $(0, -2, 0, 2, 0, 0)^\top$
iii. $(0, 0, 0, 0, 0, 2)^\top$

It's $(0, 0, 0, 0, 0, 2)^\top$. A basic solution has at most two non-zero entries (since the matrix A has rank at most 2) and a feasible solution has no negative entries. For $(0, 0, 0, 0, 0, 2)^\top$, different bases work, as long as they include column 6.

(b) 2 points – How would you model the constraint "x is at least the maximum of y and z" in an LP? (Here, x, y, z are variables.)

$x \geq y$ and $x \geq z$.

(c) 5 points – Let (P) be the following LP.

$$\min\{c^T x \ : \ Ax \geq b, \ x \geq \mathbb{0}\}.$$

Which of the following conditions implies that (P) is unbounded? (Select one.)

i. There is a feasible solution x and a vector d such that $d \geq \mathbb{0}$, $Ad \geq \mathbb{0}$, and $c^\top d > 0$.
ii. There is a feasible solution x and a vector d such that $d \geq \mathbb{0}, d^\top A \geq \mathbb{0}$, and $c^\top d > 0$.
iii. There is a feasible solution x and a vector d such that $d \leq \mathbb{0}$, $Ad \leq \mathbb{0}$, and $c^\top d > 0$.

Recall that a minimization problem (P) is unbounded if for every real number α, there is a feasible solution for (P) of value smaller than α.

It's iii. For every $t \geq 0$, the vector $x - td$ is feasible, and has objective value $c^\top x - tc^\top d$, which can be made arbitrarily small by increasing t.

(d) 5 points – Give an optimum solution to the LP $\max\{c^T x : Ax = b, x \geq \mathbb{0}\}$ where:

$$c = \begin{pmatrix} 5 \\ 4 \\ 0 \\ 0 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$$

If we start the simplex with basis $\{3, 4\}$, the entering variable is x_1, and the variable leaving is x_4. The simplex tableau after this looks like this: $\begin{pmatrix} c: & 1 & | & 0 & 1 & 0 & 5 & | & 30 \\ A: & 0 & | & 0 & 1 & 1 & -1 & | & 2 \\ A: & 0 & | & 1 & 1 & 0 & 1 & | & 6 \end{pmatrix}$ Since all the entries in the top row are non-negative, this is an optimum solution: $(6, 0, 2, 0)^\top$. It has value 30.

2) You are going on holiday! On your flight back, you can fill your entire suitcase with gifts; but the airline is very strict that your suitcase must not exceed 18 kg in weight, and also must not exceed 65 litres in volume. The total cost of items you bring also should not exceed 800$.

Gift item	Weight (kg)	Volume (L)	Cost ($)	Happiness points
Candy	10	5	10	10
Artwork	2	15	100	20
Trinkets	3	5	25	5

Resource limits:

- Total weight: at most 18 kg
- Total volume: at most 65 L
- Total cost: at most 800$

(a) 6 points – Write an LP to maximize total happiness points subject to these constraints. (Assume that gift items can be purchased in fractional quantities; you cannot bring negative quantities, however.)

Let c, a, t denote the amount of candy, artwork, and trinkets we bring home. Then we have:

$$\begin{aligned} \max\ \ & 10c + 20a + 5t \\ s.t.\ & 10c + 2a + 3t \leq 18 \\ & 5c + 15a + 5t \leq 65 \\ & 10c + 100a + 25t \leq 800 \\ & c, a, t \geq 0. \end{aligned}$$

The objective function reflects the total happiness points, while the first three rows model the constraints on weight, volume, and cost, respectively. Finally, the last row encodes that all quantities should be non-negative.

(b) 6 points – Which constraint can we omit without changing the optimum solution of the LP? Justify your answer.

Cost! If we look at the constraints, then because of the weight constraint, we can buy at most two units of candy, and at most 9 units of trinkets. Because of the volume constraint, we can buy at most 5 units of artwork. In other words, we always have $c \leq 2$ and $t \leq 9$ and $a \leq 5$. But that implies that cost is as most $20 + 500 + 225 = 745 < 800$ for any solution satisfying the weight and volume constraints.
We actually can omit $c \geq 0$ or $a \geq 0$ without any changes to the optimum solution. However, this is harder to justify. If we omit $t \geq 0$, then the LP becomes unbounded. One ray is $(q, q, -5q)$.

3) Consider the LP (P) given by $\max\{c^T x : Ax = b, x \geq \mathbb{0}\}$ where $a \in \mathbb{R}$ is a positive real number and:

$$c = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 1 & 4 \\ 3 & -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ a \end{pmatrix}$$

(a) 3 points – Write down the auxiliary LP for solving (P) using Two-Phase Simplex. You do not need to put it in standard form.

It is $\min\{c^\top x : Ax = b, x \geq \mathbb{0}\}$ where

$$c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 1 & 4 & 1 & 0 \\ 3 & -1 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ a \end{pmatrix}$$

(b) 5 points – For which values of a is (P) infeasible? Give the corresponding certificate(s). *Hint: You might recall the formula $A_B^{-\top} c_B$.*

When $a \leq 6$, there is a feasible solution $(a/3, 0, (4 - 2a/3)/4)^\top$. Easy calculations show that this is feasible so long as $a \leq 6$ (using that we also assumed $a > 0$). So we are looking for a vector $y = (y_1, y_2)^\top$ such that $y^\top A \geq \mathbb{0}$ and $y^\top b < 0$ when $a > 6$. That gives us a clue: Maybe we should pick $y_1 = 1$ and $y_2 = -2/3$. I picked this because then $y^\top b < 0$ if and only if $a > 6$. Let's compute $y^\top A = (0, 5/3, 4)$. That works, so we have our certificate of infeasibility for all $a > 6$. Note that any non-zero multiple of y also works; though to match the format from class, it should be a positive multiple.

(c) 3 points – For which values of a is (P) unbounded? Give the corresponding certificate(s). You should also justify why (P) is bounded for the remaining values of a.

Never. We can see that x_3 is irrelevant to the objective value. Also, since x_2 has coefficient -1, that means that the most x_2 can contribute to the objective value is 0. Finally, since $2x_1 + x_2 + 4x_3 = 4$ (by the first constraint) and since $x_2, x_3 \geq 0$, we have $x_1 \leq 2$. Therefore, for every feasible solution, its value is at most 2.

(d) 10 points – For which values of a does (P) have an optimum solution? Give the corresponding optimum solution(s). *Hint: Either use the simplex algorithm or consider all basic feasible solutions.*

When $a \leq 6$, by the fundamental theorem of LPs. There are at most three basic feasible solutions, since there are at most three bases: $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$. We can see that $\{2, 3\}$ is not a feasible basis, because the second constraint would say $-x_2 = a$, but $a > 0$, and so $x_2 < 0$, a contradiction. So we only need to look at the other two options. For $\{1, 3\}$, we actually computed the BFS above:

$$\left(\frac{a}{3}, 0, \frac{4 - 2a/3}{4}\right) = \left(\frac{a}{3}, 0, \frac{6 - a}{6}\right),$$

so let's compute the BFS for $\{1, 2\}$:

$$\left(\frac{4 + a}{5}, \frac{12 - 2a}{5}, 0\right).$$

These are clearly BFS since $a \leq 6$. We need to compare their objective values.

$$\frac{4 + a}{5} - \frac{12 - 2a}{5} = \frac{3a - 8}{5}.$$

When is

$$(3a - 8)/5 > a/3$$

true? When $9a - 24 > 5a$ so then $4a > 24$, which is impossible when $a \leq 6$. So the optimum solution is

$$\left(\frac{a}{3}, 0, \frac{4 - 2a/3}{4}\right) = \left(\frac{a}{3}, 0, \frac{6 - a}{6}\right).$$

You may use this page for rough work or to continue any other question that you have run out of space answering.

If you use this page to continue a question, **you must** indicate clearly, in the original location, that the work continues here, and indicate on this page which question you are continuing. **If you fail to do so, the work will not be graded.** You may tear off this page but <u>you must include this page when you submit your exam.</u>

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